

CSC/SDS 109: Communicating with Data

a2: Visual Encoding Basics Check

Goals:

- Decompose a visualization into marks and channels
- Compose a visualization

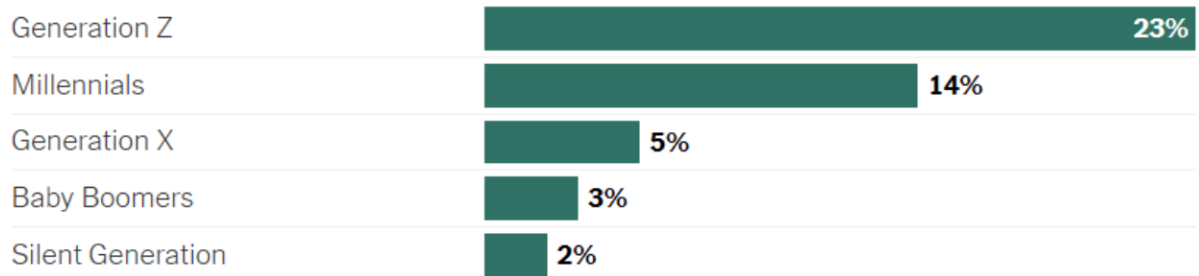
Instructions

Visual encodings (visualizations) are composed of marks and channels. For each of the visual encodings below, identify the mark(s) and what one mark represents. Then, fill out this table:

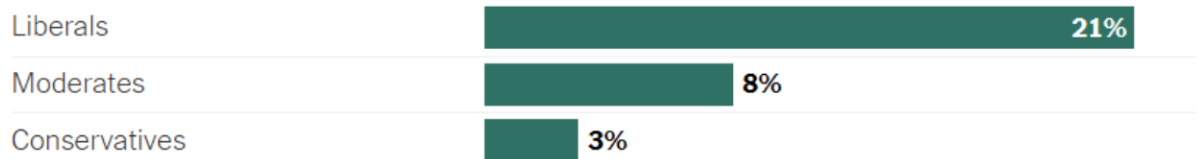
Variable	Variable Type	Visual Channel	Visual Channel Type	Do the variable type and visual channel type match?
.	.	.	.	.
.	.	.	.	.
.	.	.	.	.

L.G.B.T.Q. identification by...

Age



Ideology



Locale



Education



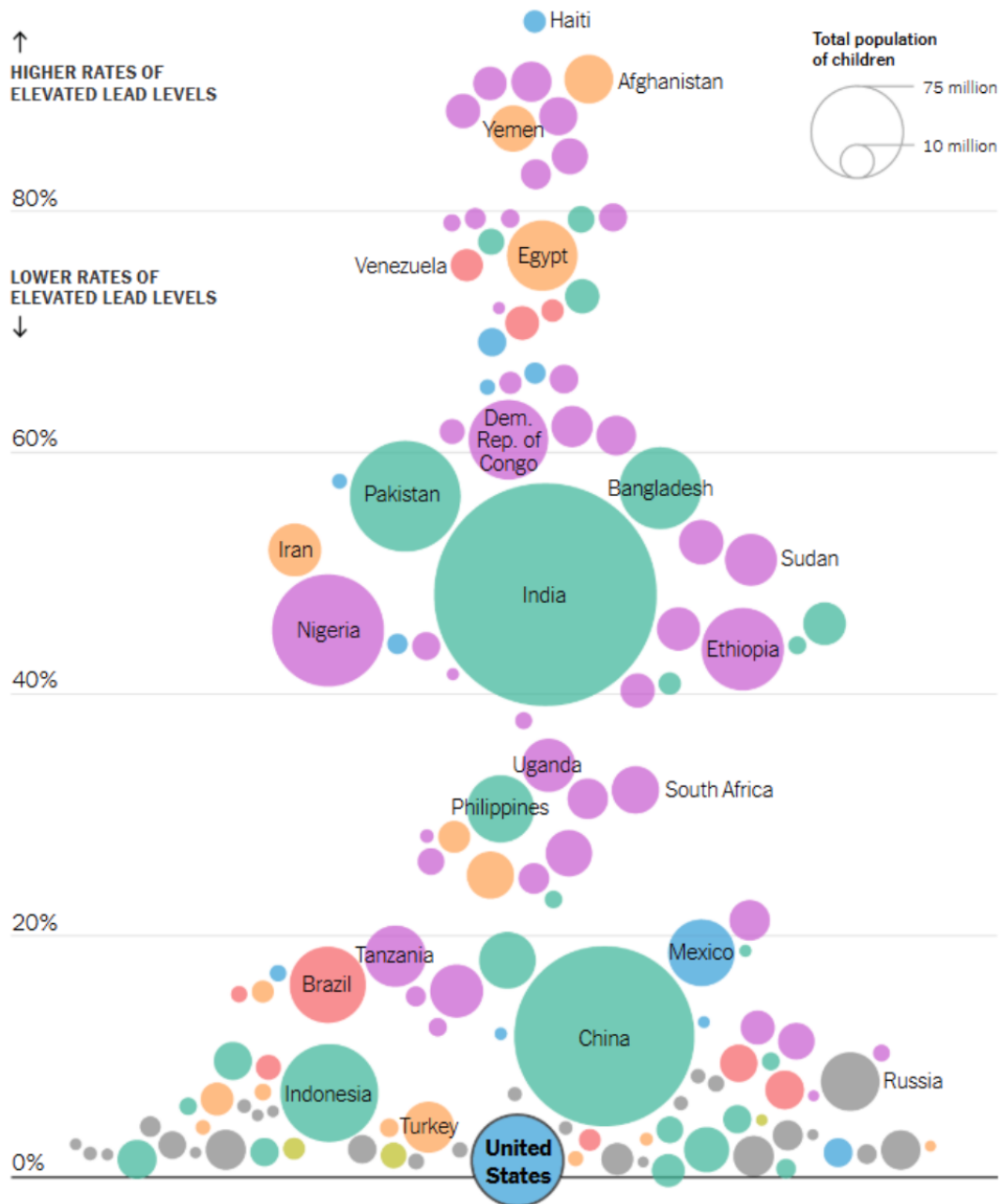
Figures are rounded - Source: Gallup

Source: <https://www.nytimes.com/2025/04/24/learning/whats-going-on-in-this-graph-april-30-2025.html>

Estimated share of children with blood lead levels at or above 5 micrograms per deciliter

AFRICA ASIA EUROPE MIDDLE EAST NORTH AMERICA OCEANIA SOUTH AMERICA

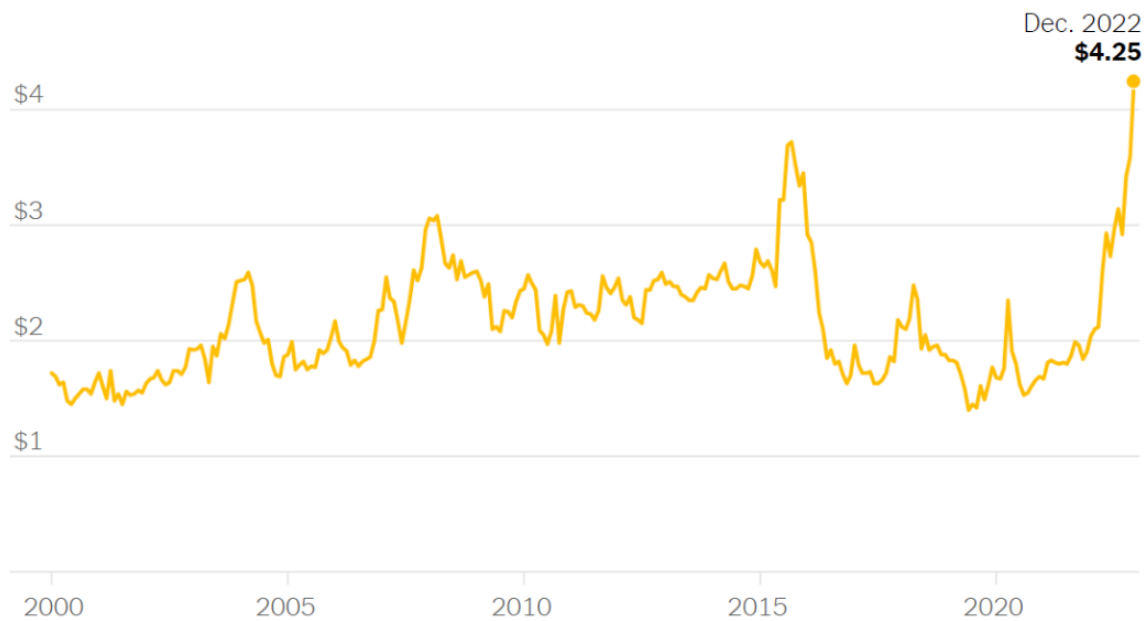
>99% of children with high lead levels



Sources: Institute for Health Metrics and Evaluation, Pure Earth, UNICEF - Note: Country estimates are based on limited data and may vary. Data for most countries is from 2021.

Source: <https://www.nytimes.com/2025/01/16/learning/whats-going-on-in-this-graph-jan-29-2025.html>

Monthly average price of a dozen eggs



Note: Prices are in December 2022 dollars. • Source: Bureau of Labor Statistics

Source: <https://www.nytimes.com/2023/03/30/learning/whats-going-on-in-this-graph-april-5-2023.html>

For each of the following datasets and tasks, draw an appropriate visual encoding. All data comes from the common dataset: <https://www.smith.edu/your-campus/offices-services/institutional-research/data-about-smith>

(1) Draw a visualization that compares undergraduate demographics from 2010-11 to 2025-26.

Demographics	2025-26	2010-11
American Indian or Alaska Native	<1%	<1%
Asian	9.5%	11.2%
Black or African American	5.7%	5.3%
Latine	13.8%	7.3%
Native Hawaiian or Other Pacific Islander	<1%	<1%
Two or More Races	6.9%	3.4%
Unknown	2.7%	20.1%
White	48.5%	43.3%
International	12.8%	8.9%

(2) Draw a visualization that shows how admission rate changes over time.

Academic Year	Admit Rate
2021-22	30%
2022-23	23%
2023-24	20%
2024-25	21%
2025-26	22%

(3) Choose data from the common dataset and a task of interest to you and draw a visualization:

<https://drive.google.com/drive/folders/1yhtsw9GRf0e2yItkGXFJ6XuqzZxuN0Db>

Competency Demonstration

This is a competency-based course. After you complete your assignment, write a short, succinct, and clear reflection explaining which core competencies you demonstrated in your work and how.

It is recommended that you use this assignment to demonstrate any of the following competencies:

- Map data to appropriate visual channels

Submission

Submit your completed work as a PDF on Gradescope.